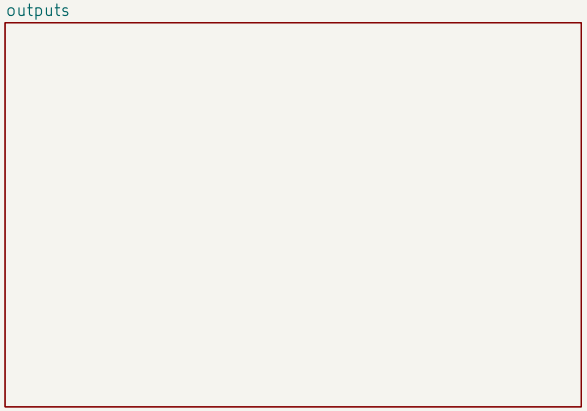
power



Файл: power.kicad_sch

outputs



Файл: outputs.kicad_sch

mcu



Файл: mcu.kicad_sch

comms

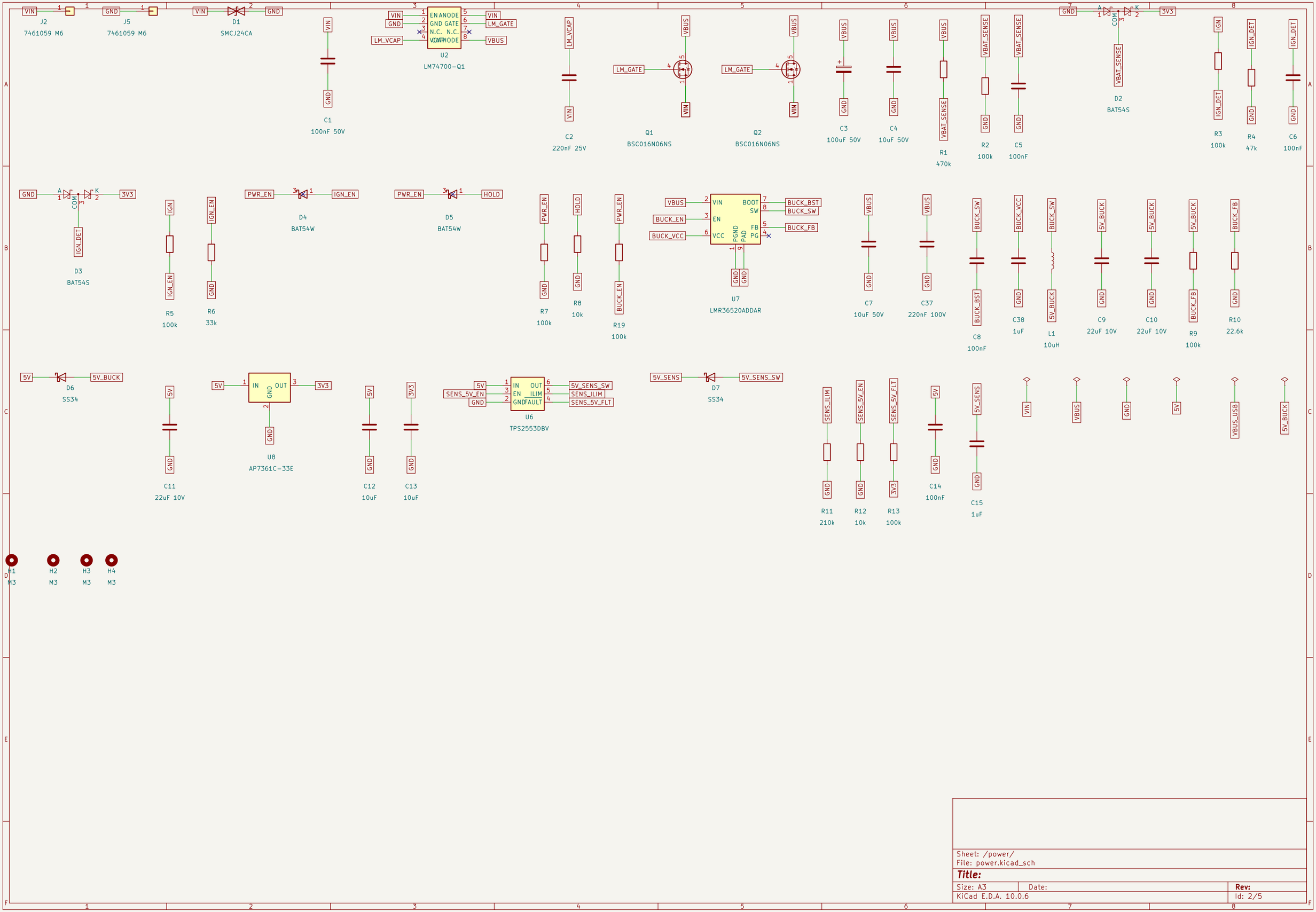


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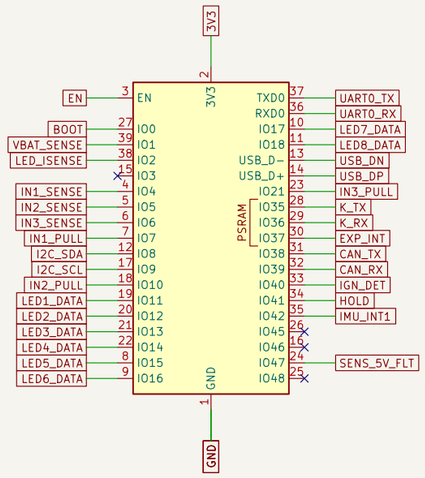
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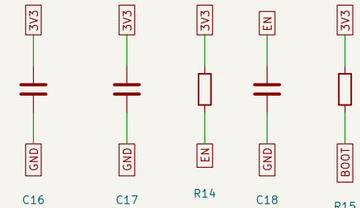
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R20
10k
GND
LED1_IN
R21
4.7k
LED1_IN_R
GND
LED2_IN
R22
10k
GND
LED2_IN
R23
4.7k
LED2_IN_R
GND
LED3_IN
R24
10k
GND
LED3_IN
R25
4.7k
LED3_IN_R
GND
LED4_IN
R26
10k
GND
LED4_IN
R27
4.7k
LED4_IN_R
GND
LED5_IN
R28
10k
GND
LED5_IN
R29
4.7k
LED5_IN_R
GND
LED6_IN
R30
10k
GND
LED6_IN
R31
4.7k
LED6_IN_R
GND
LED7_IN
R32
10k
GND
LED7_IN
R33
4.7k
LED7_IN_R
GND
LED8_IN
R34
10k
GND
LED8_IN
R35
4.7k
LED8_IN_R
DSEL
R36
10k
GND
VBUS
LED1_IN_R
LED2_IN_R
DEN1_R
DSEL_R1
LED_ISENSE_RAW
GND
VS
IN0
IN1
DEN
DSEL
IS
GND
NC
OUT0
OUT0
OUT1
OUT1
OUT1
NC
LED1_12V
LED1_12V
LED1_12V
LED2_12V
LED2_12V
LED2_12V
U10
BTS7008-2EPA
DEN1
R37
10k
GND
DEN1
R38
4.7k
DEN1_R
DSEL
R39
4.7k
DSEL_R1
VBUS
C20
100nF
GND
LED1_12V
C21
10nF 50V
GND
LED2_12V
C22
10nF 50V
GND
VBUS
LED_ISENSE_RAW
LED3_IN_R
LED4_IN_R
DEN2_R
DSEL_R2
GND
LED3_12V
LED3_12V
LED3_12V
LED4_12V
LED4_12V
LED4_12V
U11
BTS7008-2EPA
DEN2
R40
10k
GND
DEN2
R41
4.7k
DEN2_R
DSEL
R42
4.7k
DSEL_R2
VBUS
C23
100nF
GND
LED3_12V
C24
10nF 50V
GND
LED4_12V
C25
10nF 50V
GND
VBUS
LED_ISENSE_RAW
LED5_IN_R
LED6_IN_R
DEN3_R
DSEL_R3
GND
LED5_12V
LED5_12V
LED5_12V
LED6_12V
LED6_12V
LED6_12V
U12
BTS7008-2EPA
DEN3
R43
10k
GND
DEN3
R44
4.7k
DEN3_R
DSEL
R45
4.7k
DSEL_R3
VBUS
C26
100nF
GND
LED5_12V
C27
10nF 50V
GND
LED6_12V
C28
10nF 50V
GND
VBUS
LED_ISENSE_RAW
LED7_IN_R
LED8_IN_R
DEN4_R
DSEL_R4
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LED7_12V
LED7_12V
LED8_12V
LED8_12V
LED8_12V
U13
BTS7008-2EPA
DEN4
R46
10k
GND
DEN4
R47
4.7k
DEN4_R
DSEL
R48
4.7k
DSEL_R4
VBUS
C29
100nF
GND
LED7_12V
C30
10nF 50V
GND
LED8_12V
C31
10nF 50V
GND
LED_ISENSE_RAW
R49
1.2k
GND
LED_ISENSE_RAW
R50
4.7k
LED_ISENSE
LED_ISENSE
C32
10nF
GND
GND
A
COM
K
3V3
LED_ISENSE
D10
BAT54S
3V3
I2C_SCL
I2C_SDA
EXP_INT
SCL
SDA
INT
VDD
VSS
IO0_0
IO0_1
IO0_2
IO0_3
IO0_4
IO0_5
IO0_6
IO0_7
IO1_0
IO1_1
IO1_2
IO1_3
IO1_4
IO1_5
IO1_6
IO1_7
LED1_IN
LED2_IN
LED3_IN
LED4_IN
LED5_IN
LED6_IN
LED7_IN
LED8_IN
DEN1
DEN2
DEN3
DEN4
DSEL
CAN_S
STATUS_LED
SENS_5V_EN
GND
GND
GND
A0
A1
A2
GND
U14
PCA9555PW
3V3
C33
100nF
GND
EXP_INT
R51
10k
3V3
I2C_SDA
R52
4.7k
3V3
I2C_SCL
R53
4.7k
3V3
5V
LED1_DATA
LED2_DATA
LED3_DATA
LED4_DATA
GND
GND
GND
GND
5V
GND
A0
A1
A2
A3
A4
A5
A6
A7
VCC
A->B
CE
B0
B1
B2
B3
B4
B5
B6
B7
GND
LED1_BUF
LED2_BUF
LED3_BUF
LED4_BUF
GND
GND
U15
74HCT245PW
5V
C34
100nF
GND
5V
LED5_DATA
LED6_DATA
LED7_DATA
LED8_DATA
GND
GND
GND
GND
5V
GND
A0
A1
A2
A3
A4
A5
A6
A7
VCC
A->B
CE
B0
B1
B2
B3
B4
B5
B6
B7
GND
LED5_BUF
LED6_BUF
LED7_BUF
LED8_BUF
GND
GND
U16
74HCT245PW
5V
C35
100nF
GND
LED1_BUF
LED1_DO
R54
33
LED2_BUF
LED2_DO
R55
33
LED3_BUF
LED3_DO
R56
33
LED4_BUF
LED4_DO
R57
33
LED5_BUF
LED5_DO
R58
33
LED6_BUF
LED6_DO
R59
33
LED7_BUF
LED7_DO
R60
33
LED8_BUF
LED8_DO
R61
33
LED1_DO
GND
D11
PESD5V0S1BA
LED2_DO
GND
D12
PESD5V0S1BA
LED3_DO
GND
D13
PESD5V0S1BA
LED4_DO
GND
D14
PESD5V0S1BA
LED5_DO
GND
D15
PESD5V0S1BA
LED6_DO
GND
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GND
D17
PESD5V0S1BA
LED8_DO
GND
D18
PESD5V0S1BA

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U1

ESP32-S3-WROOM-1U-N16R2



C16 22uF 10V

C17 100nF

R14 10k

C18 1uF

R15 10k



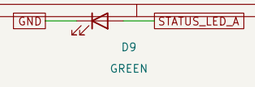
SW1

BOOT



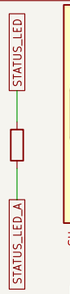
SW2

EN



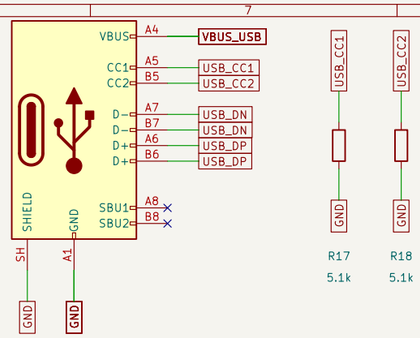
D9

GREEN



R16

1k

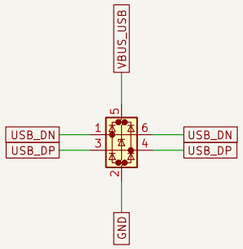


J3

USB-C 16P

R17 5.1k

R18 5.1k



U9

USBLC6-2SC6



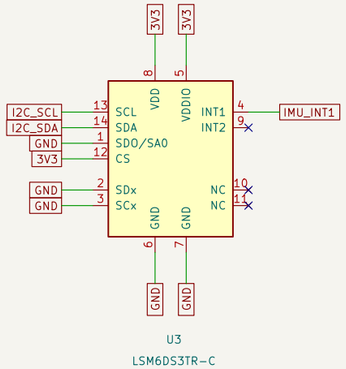
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SS34



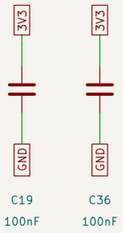
J4

UART0



U3

LSM6DS3TR-C



C19 100nF

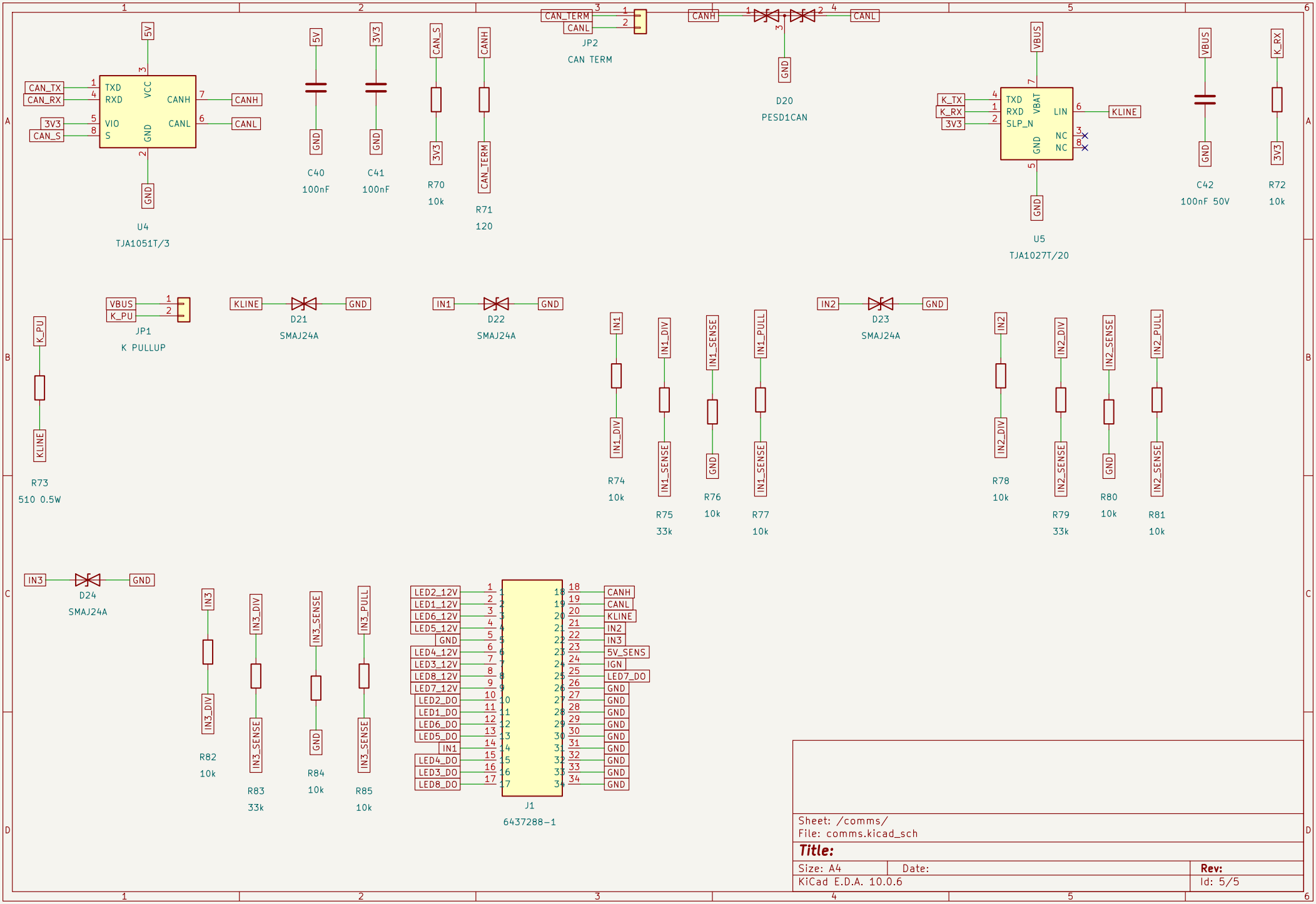
C36 100nF

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5V
CAN_TX
CAN_RX
3V3
CAN_S
TXD
RXD
VIO
S
VCC
GND
CANH
CANL
U4
TJA1051T/3
C40
100nF
C41
100nF
R70
10k
R71
120
CAN_TERM
JP2
CAN TERM
D20
PESD1CAN
VBUS
K_TX
K_RX
SLP_N
VBAT
LIN
NC
KLINE
U5
TJA1027T/20
C42
100nF 50V
K_RX
R72
10k
K_PU
R73
510 0.5W
JP1
K PULLUP
D21
SMAJ24A
D22
SMAJ24A
IN1
IN1_DIV
IN1_SENSE
IN1_PULL
R74
10k
R75
33k
R76
10k
R77
10k
D23
SMAJ24A
IN2
IN2_DIV
IN2_SENSE
IN2_PULL
R78
10k
R79
33k
R80
10k
R81
10k
IN3
D24
SMAJ24A
IN3_DIV
IN3_SENSE
IN3_PULL
R82
10k
R83
33k
R84
10k
R85
10k
LED2_12V
LED1_12V
LED6_12V
LED5_12V
LED4_12V
LED3_12V
LED8_12V
LED7_12V
LED2_DO
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LED6_DO
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LED3_DO
LED8_DO
5V_SENS
IGN
LED7_DO
J1
6437288-1
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